

Data Mining and Visualization Project

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Presenting Data:

Generally, we have 27,213 records:

```
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   ID                    27213 non-null  object
1   weekend_nights         27213 non-null  int64
2   week_nights           27213 non-null  int64
3   room_type             27213 non-null  object
4   board_type            19045 non-null  object
5   n_adults              27213 non-null  int64
6   n_less_12            27213 non-null  int64
7   n_more_12            27213 non-null  int64
8   booked_tour           27213 non-null  int64
9   n_requests            27213 non-null  int64
10  lead_time             26794 non-null  float64
11  purchase_type         22366 non-null  object
12  n_p_cacellation       27213 non-null  int64
13  n_p_not_cacellation   27213 non-null  int64
14  repeated              27213 non-null  int64
15  price                 23808 non-null  float64
16  date                  27213 non-null  object
17  is_canceled           27213 non-null  int64
dtypes: float64(2), int64(11), object(5)
```

Which I chose to divide as such:

```
# Numerical Columns
numerical_columns = ['n_adults', 'n_less_12', 'n_more_12', 'weekend_nights', 'week_nights', 'lead_time', 'price', 'n_requests', 'n_p_cacellation', 'n_p_not_cacellation']

# Binary Columns
binary_columns = ['booked_tour', 'repeated', 'is_canceled']

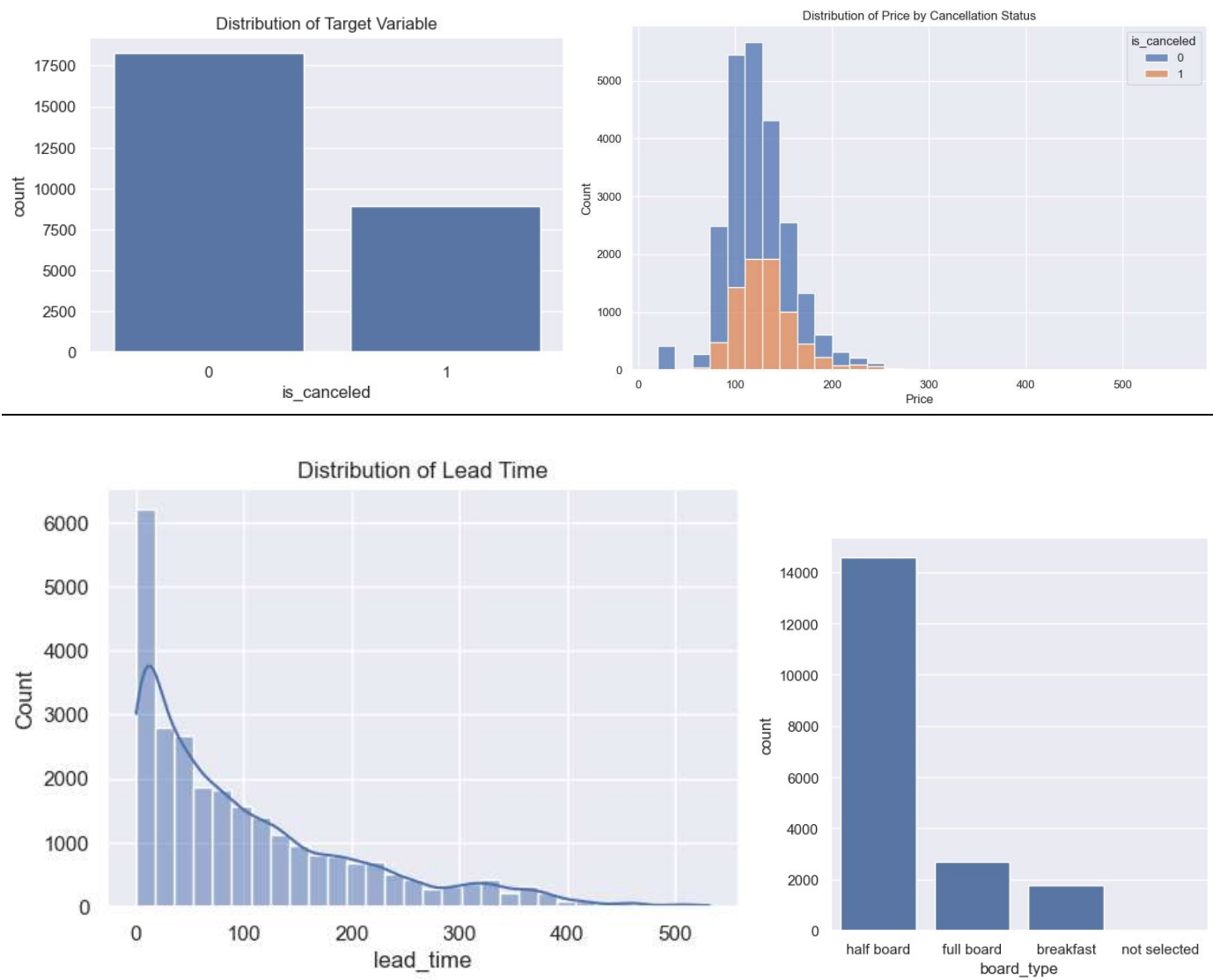
# Categorical Columns
categorical_columns = ['board_type', 'room_type', 'purchase_type']
```

Skewness:

I have checked for skewness

n_adults	-0.324161
n_less_12	5.749665
n_more_12	5.569096
weekend_nights	0.718789
week_nights	1.542295
lead_time	1.291447
price	0.682393
n_requests	1.143745
n_p_cacellation	26.069040
n_p_not_cacellation	19.076806

Attributes distribution:



Correlation between attributes:

Note: before checking for correlation I encoded the categorical attributes to numerical in a copy called : 'vis_df'

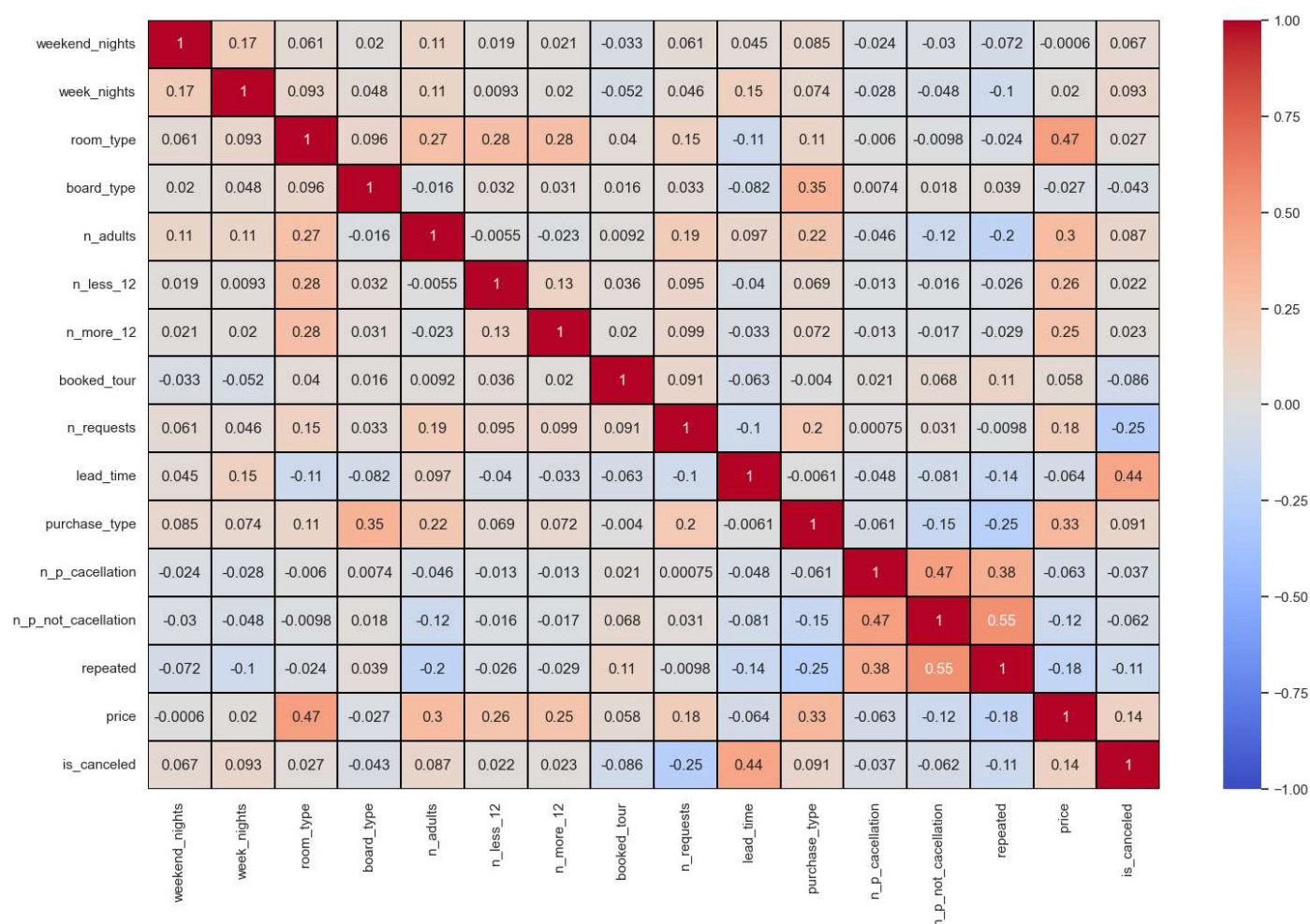
Missing Values (Null Values):

When searching for null values I found:

```
ID          0
weekend_nights  0
week_nights    0
room_type      0
board_type    8168
n_adults       0
n_less_12      0
n_more_12      0
booked_tour    0
n_requests     0
lead_time     419
purchase_type  4847
n_p_cacellation  0
n_p_not_cacellation  0
repeated      0
price         3405
date          0
is_canceled   0
dtype: int64
```

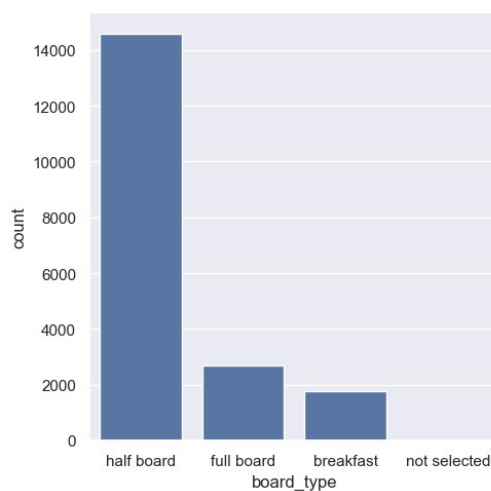
Missing Values:

1. board_type (object) :missing 8168, about 1/3 of the values are missing.
2. lead_time (float) : 419 missing
3. purchase_type (object) : missing 4847
4. price (float) : missing 3405

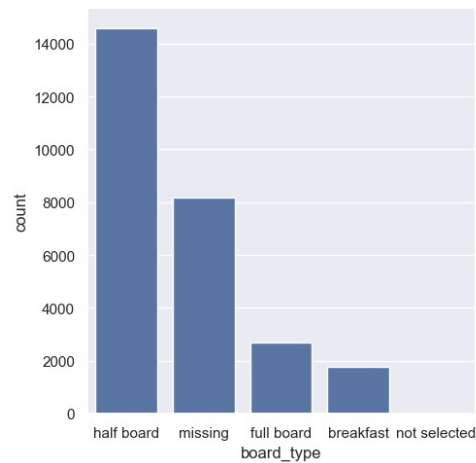


Using the correlation matrix and the distributions I will now show I decided to preform the following:

1. Board Type:



This attribute has many missing values and changing them to an exiting category might affect the outcomes of classification models, for this, I decided to fill null values with 'missing'



2. **Lead Time:**

Small amount of values, because of the variance in the data, I choose to fill the null values with the median.

3. **Purchase Type:**

Similarly to board type, I decided to fill the null values with 'missing'

4. **Price:**

Price had high correlation to room type. So I decided to fill the null values based on the room type. For room types that had few missing values I used the mean price, for room types that had many, I used normally distributed values.

Outliers:

I took a look into the numerical and binary attributes and found no outliers.

```
board_type
half board    14591
missing       8168
full board    2697
breakfast     1753
not selected    4
Name: count, dtype: int64
room_type
Room_Type 1    21084
Room_Type 4    4571
Room_Type 6     741
Room_Type 2     511
Room_Type 5     180
Room_Type 7     123
Room_Type 3        3
Name: count, dtype: int64
```

For the categorical attributes I found:

1. Board Type: only 4 values 'not selected', changed to 'missing'.
2. Room Type: only 3 values for "Room_Type 3", changed to other room types based on price.

Adding attributes:

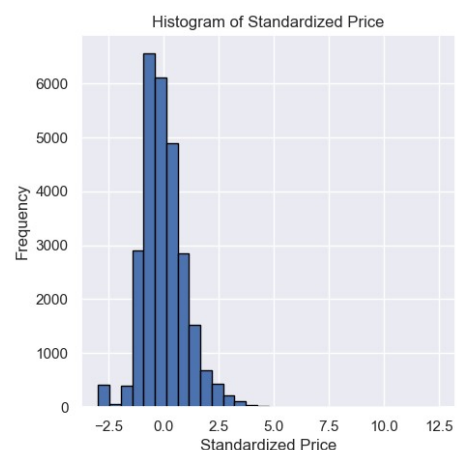
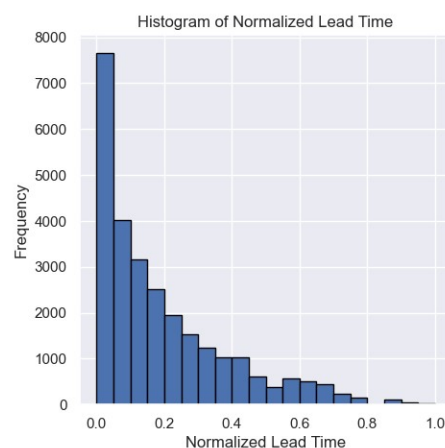
I chose to add an attribute 'length' which is the total amount of nights (midweek + weekend).

Inconsistency:

I looked into: unique IDs, spelling of categories and found no inconsistencies.

Normalization and Discretization:

I have two major continuous attributes: 'price' and 'lead_time'. Initially, considered discretizing 'price' into 5 equal-width bins representing different price levels. However, I noticed that most values fall into the first two levels, so I considered normalizing 'price' using a z-score instead. Also, I decided to normalize 'lead_time'.



Data Reduction:

I considered using PCA but after I saw the results I decided I shouldn't use it because the dataset does not have many features compared to other databases where reduction is mandatory, but PCA also did not reduce any dimensions. Moreover, the PCA output is not interpretable for further analysis.

Imbalanced Data:

Examining the distribution of our data based on the target value reveals a 2:1 ratio of non-canceled to canceled records, indicating the data is imbalanced but not highly so. To address this, I will apply Cost-Sensitive Learning in my models by setting 'class_weight = balanced'. Another approach considered but not adopted after comparisons was using algorithms that support cost-sensitive learning by balancing the data with over/down sampling algorithms and turning off 'class_weight = balanced'.

Classifiers:

Brief:

- **Choosing Evaluation Metrics:** I selected the most appropriate evaluation metrics for the problem.
- **Data Splitting and Cross-Validation:** I split the data into train and validation sets and used cross-validation.
- **Hyperparameter Tuning:** I performed hyperparameter tuning for each model, explaining the choice of parameters and their ranges.
- **Training Classifiers:** I trained at least three different classifiers, including one not covered in class, and compared them using evaluation metrics.
- **Choosing the Best Model:** I chose the best model based on statistical significance to apply to the test dataset and saved the results in a CSV file.
- **Test Data Changes:** Applied corresponding changes to the test data.

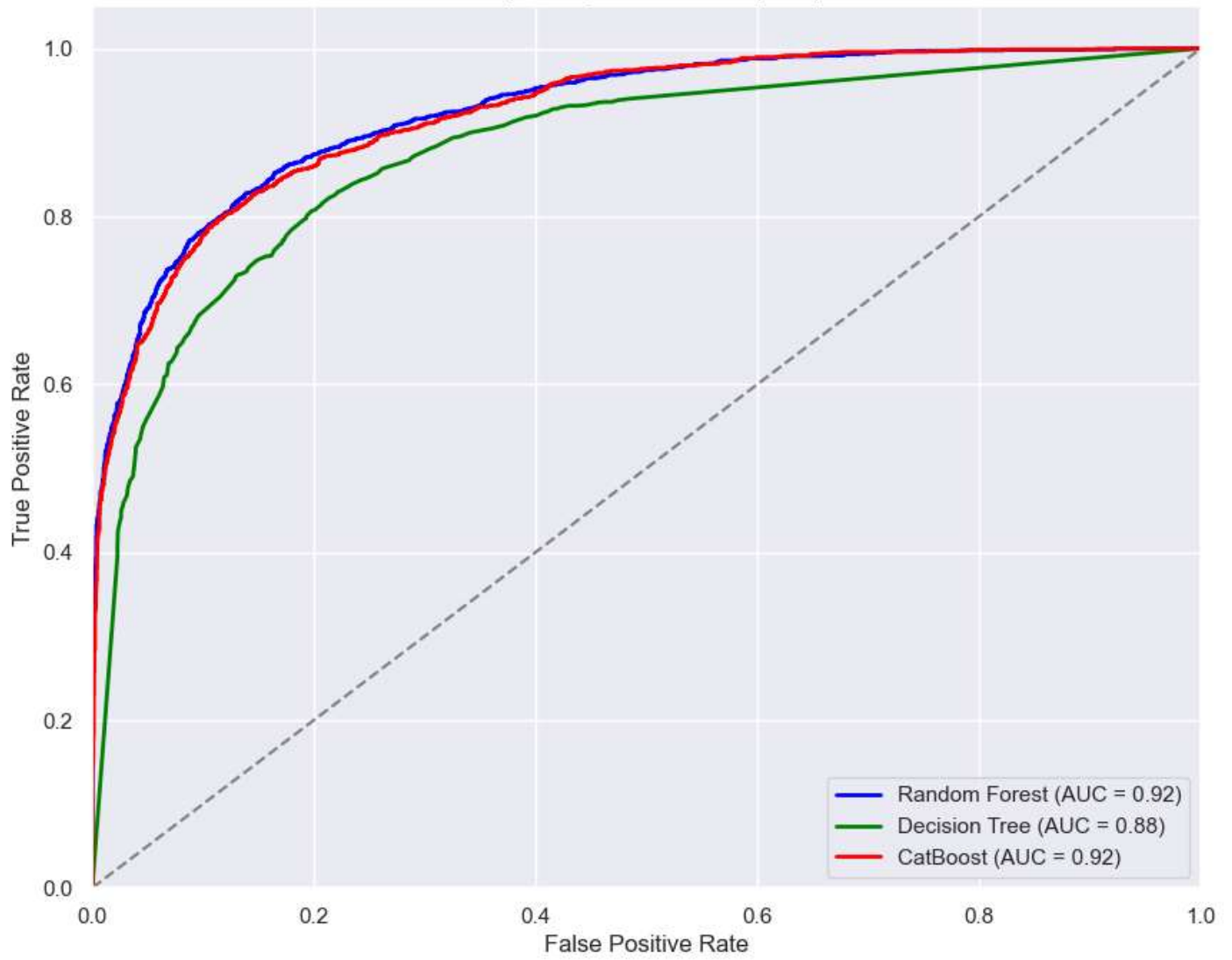
Appropriate Evaluation Metrics: In business hotel reservations, both false positives (predicting a cancellation when there isn't one) and false negatives (not predicting a cancellation when there is one) have significant implications. False positives could lead to unnecessary overbooking, while false negatives could result in revenue loss due to unoccupied rooms. Therefore, the F1-score, which balances precision and recall, is the best metric.

Selected Models: I selected Decision Tree, Random Forest, and CatBoost which wasn't covered in class. I chose Random Forest for its superior performance, ability to handle large datasets, techniques to combat overfitting, and robustness to data issues like missing values and unbalanced datasets.

Strategy:

- Optimized each model's performance by hyperparameter tuning and cross-validation.
- Evaluated the optimized models together using cross-validation and various metrics (AUC, Accuracy, Precision, Recall, F1-score), and calculated the mean values.
- Compared the models AUC.
- Estimated the preference between the classifiers using a two-sided t-test.
- Made a conclusive decision based on these evaluations.

Receiver Operating Characteristic (ROC) Curves



```

Comparing RandomForestClassifier(class_weight='balanced', max_depth=21, n_estimators=216,
                                random_state=42) and DecisionTreeClassifier(criterion='entropy', max_depth=20, max_features='sqrt',
                                min_samples_leaf=2, min_samples_split=10,
                                random_state=42):
Precision p-value: 1.3447636415486927e-07
Reject null hypothesis: There is a significant difference in precision.
Accuracy p-value: 1.5217464707889645e-07
Reject null hypothesis: There is a significant difference in accuracy.
AUC p-value: 5.238768471316712e-12
Reject null hypothesis: There is a significant difference in AUC.

Comparing RandomForestClassifier(class_weight='balanced', max_depth=21, n_estimators=216,
                                random_state=42) and <catboost.core.CatBoostClassifier object at 0x000001CE00F8BEE0>:
Precision p-value: 0.19220625011210543
Accept null hypothesis: No significant difference in precision.
Accuracy p-value: 0.19985826358547829
Accept null hypothesis: No significant difference in accuracy.
AUC p-value: 0.00012273740886443736
Reject null hypothesis: There is a significant difference in AUC.

Comparing DecisionTreeClassifier(criterion='entropy', max_depth=20, max_features='sqrt',
                                min_samples_leaf=2, min_samples_split=10,
                                random_state=42) and <catboost.core.CatBoostClassifier object at 0x000001CE00F8BEE0>:
Precision p-value: 1.8548937744130358e-07
Reject null hypothesis: There is a significant difference in precision.
Accuracy p-value: 1.6772377743639655e-07
Reject null hypothesis: There is a significant difference in accuracy.
AUC p-value: 2.181801158528634e-12
Reject null hypothesis: There is a significant difference in AUC.

```

Handling Test Set:

- Handled missing values in the same way as the training dataset.
- Applied the same normalizations as on the training dataset.
- Executed predictions using the optimized Random Forest model and generated a CSV file with the predictions (0/1) as requested.
- The prediction results are:

```

Number of positive predictions: 2350
Number of negative predictions: 6722

```